Advanced Control Systems

Assignment 3: Dynamic Modeling, Complete Equations of Motion, Christoffel Symbols, and Skew-Symmetry Analysis for a 3-DOF PRR Manipulator

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Abstract

This report presents the complete derivation and formal characterization of the dynamic equations of motion for a spatial 3-DOF PRR (Prismatic-Revolute-Revolute) robotic manipulator defined by the URDF specification (`00001_PRR_zzx FILIPPO SEGALLA.urdf`). Building upon the kinematic and dynamic energy results established in Assignment #1 and Assignment #2, the equations of motion are systematically derived in joint space using the Euler-Lagrange analytical mechanics framework: $\mathbf{B}(q)\ddot{q} + \mathbf{C}(q,\dot{q})\dot{q} + \mathbf{g}(q) = \boldsymbol{\tau}$. The Coriolis and centrifugal terms are computed component-by-component via Christoffel symbols of the first type. Furthermore, the essential structural passivity property—the skew-symmetry of the matrix $(\dot{\mathbf{B}}(q) - 2\mathbf{C}(q,\dot{q}))$—is analytically proven and numerically verified. Both parametric symbolic formulas and clean decimal numerical expressions evaluated with exact rational arithmetic are presented.

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1 System Architecture & Dynamic Framework

1.1 Description of the PRR Dynamic Model

What is being done: We formulate the inverse dynamic model of the spatial 3-DOF PRR manipulator. The mechanical parameters extracted from the URDF file and validated in prior assignments are summarized as follows:

1. **Link 1 (Prismatic Joint, q_1):** Mass $m_1 = 1.0$ kg, rectangular geometry (0.036 m $\times 0.036$ m $\times$ 0.5 m).
2. **Link 2 (Revolute Joint, q_2):** Mass $m_2 = 1.0$ kg, solid cylinder along Z_2 ($r = 0.019$ m, $h = 0.4$ m).
3. **Link 3 (Revolute Joint, q_3):** Mass $m_3 = 1.0$ kg, solid cylinder along X_3 ($r = 0.019$ m, $h = 0.4$ m).

Why it is done: Establishing the dynamic equations of motion is indispensable for model-based robot control, trajectory simulation, and joint actuator sizing. While kinematics maps motion geometry, dynamics couples joint forces and torques $\boldsymbol{\tau}$ to joint accelerations $\ddot{q}$, velocities $\dot{q}$, and positions q.

1.2 Euler-Lagrange Analytical Method

What is being done: The dynamics are derived using the scalar Lagrangian function $\mathcal{L}(q, \dot{q}) = \mathcal{T}(q, \dot{q}) - \mathcal{U}(q)$, defined as the difference between total kinetic energy $\mathcal{T}$ and total potential energy $\mathcal{U}$. The fundamental Euler-Lagrange equations of motion are given by:

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i}\right) - \frac{\partial \mathcal{L}}{\partial q_i} = \tau_i, \quad i = 1, 2, 3 \tag{1}$$

Substituting $\mathcal{L} = \frac{1}{2}\dot{q}^T \mathbf{B}(q)\dot{q} - \mathcal{U}(q)$ into the variational formulation yields the step-by-step derivatives:

$$\left(\frac{\partial \mathcal{L}}{\partial \dot{q}}\right)^T = \mathbf{B}(q)\dot{q} \tag{2}$$

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}}\right)^T = \mathbf{B}(q)\ddot{q} + \dot{\mathbf{B}}(q)\dot{q} \tag{3}$$

$$\left(\frac{\partial \mathcal{L}}{\partial q}\right)^T = \frac{1}{2}\left(\frac{\partial}{\partial q}\left(\dot{q}^T \mathbf{B}(q)\dot{q}\right)\right)^T - \left(\frac{\partial \mathcal{U}}{\partial q}\right)^T \tag{4}$$

Combining terms yields the canonical matrix differential equation:

$$\mathbf{B}(q)\ddot{q} + \mathbf{C}(q, \dot{q})\dot{q} + \mathbf{g}(q) = \boldsymbol{\tau} \tag{5}$$

Why it is done: The Euler-Lagrange method avoids computing internal constraint forces between links (unlike Newton-Euler vector mechanics) and directly delivers the generalized forces $\boldsymbol{\tau}$ performing work on the generalized coordinates q.

2 Derivation of Dynamic Matrices

2.1 Generalized Inertia Matrix $\mathrm{B}(q)$ Recapitulation

What is being done: The symmetric, positive-definite inertia matrix $\mathbf{B}(q) \in \mathbb{R}^{3\times 3}$ derived in Assignment #2 is given symbolically by:

$$\mathbf{B}(q) = \begin{bmatrix} m_1 + m_2 + m_3 & 0 & \frac{m_3}{5}\cos q_3 \\ 0 & I_{2yy} + I_{3yy}\cos^2 q_3 + I_{3xx}\sin^2 q_3 + \frac{m_3}{25}\cos^2 q_3 & 0 \\ \frac{m_3}{5}\cos q_3 & 0 & I_{3zz} + \frac{m_3}{25} \end{bmatrix} \tag{6}$$

Evaluated numerically:

$$\mathbf{B}(q) = \begin{bmatrix} 3.0 & 0 & 0.2\cos q_3 \\ 0 & 0.1068 - 0.05324\sin^2 q_3 & 0 \\ 0.2\cos q_3 & 0 & 0.05342 \end{bmatrix} \tag{7}$$

Why it is done: $\mathbf{B}(q)$ captures the configuration-dependent mass and inertia properties. The diagonal element $b_{ii}(q)$ represents the effective moment of inertia at joint i when all other joints are locked, while off-diagonal elements $b_{ij}(q)$ represent inertial acceleration coupling between joint j and joint i.

2.2 Coriolis and Centrifugal Matrix $\mathbf{C}(q, \dot{q})$ via Christoffel Symbols

What is being done: To construct $\mathbf{C}(q, \dot{q})$, we compute Christoffel symbols of the first type $c_{ijk}(q)$, defined as:

$$c_{ijk}(q) = \frac{1}{2}\left(\frac{\partial b_{ij}}{\partial q_k} + \frac{\partial b_{ik}}{\partial q_j} - \frac{\partial b_{jk}}{\partial q_i}\right) \tag{8}$$

The individual entries $c_{ij}(q, \dot{q})$ of matrix $\mathbf{C}$ are computed via velocity contraction:

$$c_{ij}(q, \dot{q}) = \sum_{k=1}^{3} c_{ijk}(q)\dot{q}_k \tag{9}$$

Evaluating all 27 Christoffel symbols yields the explicit symbolic matrix:

$$\mathbf{C}(q, \dot{q}) = \begin{bmatrix} 0 & 0 & -\frac{m_3}{5}\dot{q}_3 \sin q_3 \\ 0 & \frac{25(I_{3xx}-I_{3yy})-m_3}{50}\dot{q}_3 \sin(2q_3) & \frac{25(I_{3xx}-I_{3yy})-m_3}{50}\dot{q}_2 \sin(2q_3) \\ 0 & \frac{-25(I_{3xx}-I_{3yy})+m_3}{50}\dot{q}_2 \sin(2q_3) & 0 \end{bmatrix} \tag{10}$$

Evaluating numerically with exact rational substitution yields:

$$\mathbf{C}(q, \dot{q}) = \begin{bmatrix} 0 & 0 & -0.2\dot{q}_3 \sin q_3 \\ 0 & -0.02662\dot{q}_3 \sin(2q_3) & -0.02662\dot{q}_2 \sin(2q_3) \\ 0 & 0.02662\dot{q}_2 \sin(2q_3) & 0 \end{bmatrix} \tag{11}$$

Why it is done & Physical Interpretation:

- **Centrifugal Effects ($h_{ijj}\dot{q}_j^2$):** Terms proportional to $\dot{q}_j^2$ represent centrifugal forces acting on joint i due to rotation of link j. Note $h_{iii} \equiv 0$ always.
- **Coriolis Effects ($h_{ijk}\dot{q}_j\dot{q}_k$):** Terms proportional to $\dot{q}_j\dot{q}_k$ ($j \neq k$) represent Coriolis forces arising from simultaneous motion along multiple moving axes.
- **Christoffel Symmetry:** $c_{ijk} = c_{ikj}$ guarantees consistency across velocity products.

2.3 Gravity Torque Vector $\mathbf{g}(q)$

What is being done: The gravity force vector $\mathbf{g}(q) \in \mathbb{R}^3$ is derived as the gradient of potential energy $\mathcal{U}(q)$:

$$g_i(q) = \frac{\partial \mathcal{U}(q)}{\partial q_i} \implies \mathbf{g}(q) = \begin{bmatrix} g(m_1 + m_2 + m_3) \\ 0 \\ \frac{gm_3}{5}\cos q_3 \end{bmatrix} \tag{12}$$

Substituting $g = 9.81$ m/s^2 and link masses $m_i = 1$ kg:

$$\mathbf{g}(q) = \begin{bmatrix} 29.43 \\ 0 \\ 1.962 \cos q_3 \end{bmatrix} \tag{13}$$

Why it is done: The gravity vector specifies the static holding forces/torques required to maintain equilibrium ($\ddot{q} = \mathbf{0}, \dot{q} = \mathbf{0}$). Joint 2 requires zero gravity torque ($g_2 = 0$) because rotation about Z_0 occurs purely in a horizontal plane.

3 Passivity & Skew-Symmetry Analysis

3.1 Matrix Derivative $\dot{\mathrm{B}}(q)$ & Skew-Symmetry Proof

What is being done: We compute the time derivative of the generalized inertia matrix $\dot{\mathbf{B}}(q) = \sum_{k=1}^{3} \frac{\partial \mathbf{B}}{\partial q_k} \dot{q}_k$:

$$\dot{\mathbf{B}}(q) = \begin{bmatrix} 0 & 0 & -\frac{m_3}{5}\dot{q}_3 \sin q_3 \\ 0 & \frac{25(I_{3xx}-I_{3yy})-m_3}{25}\dot{q}_3 \sin(2q_3) & 0 \\ -\frac{m_3}{5}\dot{q}_3 \sin q_3 & 0 & 0 \end{bmatrix} \tag{14}$$

We then evaluate the matrix $\mathbf{N}(q, \dot{q}) = \dot{\mathbf{B}}(q) - 2\mathbf{C}(q, \dot{q})$:

$$\mathbf{N}(q, \dot{q}) = \begin{bmatrix} 0 & 0 & \frac{m_3}{5}\dot{q}_3 \sin q_3 \\ 0 & 0 & \frac{-25(I_{3xx}-I_{3yy})+m_3}{25}\dot{q}_2 \sin(2q_3) \\ -\frac{m_3}{5}\dot{q}_3 \sin q_3 & \frac{25(I_{3xx}-I_{3yy})-m_3}{25}\dot{q}_2 \sin(2q_3) & 0 \end{bmatrix} \tag{15}$$

Checking transpose equality confirms $\mathbf{N}^T = -\mathbf{N}$, meaning $\mathbf{N} + \mathbf{N}^T = \mathbf{0}_{3\times 3}$.

Why it is done & Control Implications: Skew-symmetry of $(\dot{\mathbf{B}} - 2\mathbf{C})$ implies that for any vector $\mathbf{x} \in \mathbb{R}^3$:

$$\mathbf{x}^T \left(\dot{\mathbf{B}}(q) - 2\mathbf{C}(q, \dot{q}) \right) \mathbf{x} = 0 \tag{16}$$

This fundamental property reflects conservation of energy: internal Coriolis and centrifugal forces perform zero net work on the manipulator. It is heavily exploited in constructing Lyapunov candidates for adaptive control, computed-torque control, and passivity-based control.

4 Explicit Dynamic Equations of Motion

Combining all terms into scalar equations $\tau_i = \sum_{j=1}^{3} b_{ij}\ddot{q}_j + \sum_{j=1}^{3} c_{ij}\dot{q}_j + g_i$ gives:

4.1 Scalar Dynamic Equations

$$\tau_1 = 3.0\,\ddot{q}_1 + 0.2\cos(q_3)\,\ddot{q}_3 - 0.2\sin(q_3)\,\dot{q}_3^2 + 29.43 \tag{17}$$

$$\tau_2 = \left(0.1068 - 0.05324\sin^2 q_3\right)\ddot{q}_2 - 0.05324\sin(2q_3)\,\dot{q}_2\dot{q}_3 \tag{18}$$

$$\tau_3 = 0.05342\,\ddot{q}_3 + 0.2\cos(q_3)\,\ddot{q}_1 + 0.02662\sin(2q_3)\,\dot{q}_2^2 + 1.962\cos(q_3) \tag{19}$$

4.2 Extension to General Dynamic Model

If joint viscous friction $F_v\dot{q}$, Coulomb friction $F_s\mathrm{sgn}(\dot{q})$, or an external environment wrench h_e are present, the general dynamic equation expands to:

$$\mathbf{B}(q)\ddot{q} + \mathbf{C}(q, \dot{q})\dot{q} + F_v\dot{q} + F_s\mathrm{sgn}(\dot{q}) + \mathbf{g}(q) = \boldsymbol{\tau} - \mathbf{J}^T(q)h_e \tag{20}$$